

FitBuddy

AI Fitness Plan Generator using Gemini Models

Project Analysis & Development Phases

From brainstorming and requirements to development, testing, documentation and demonstration

Presented By

J. SURENDIRAKUMAR

R. DEEPAKKUMAR

P. SAKTHIVEL

S. HARIDEV

S.E. KOWSHIK

FastAPI • Gemini 1.5 Pro • Gemini Flash • SQLite • SQLAlchemy • Jinja2

Project Overview

What FitBuddy does and how the application is structured

0

Core Purpose

Generate personalized fitness support

- Accepts age, weight, fitness goal and workout intensity.
- Generates a customized 7-day workout plan.
- Provides nutrition or recovery guidance.
- Allows users to submit feedback and regenerate the plan.

0

Technical Foundation

Modular web application

- Frontend: HTML + Jinja2 templates.
- Backend: FastAPI for routing and application logic.
- AI layer: Gemini Pro for workout plans and Gemini Flash for quick tips.
- Database: SQLite with SQLAlchemy ORM.

Project Analysis Phases

The presentation follows the eight phases requested

1

Brainstorming & Ideation

2

Requirement Analysis

3

Project Design

4

Project Planning

5

Project Development

6

Project Testing

7

Project Documentation

8

Project Demonstration

Brainstorming & Ideation Phase

Identify the problem and define the AI fitness solution.

1

Phase Objective

Identify the problem and define the AI fitness solution.

- Problem: users need structured fitness plans tailored to individual goals.
- Idea: build a web application that generates personalized workout plans using generative AI.
- Key scenarios: initial plan generation, feedback-based updates, nutrition/recovery tips, and admin monitoring.
- Target outcome: a simple and adaptive fitness-planning experience.

Requirement Analysis Phase

Translate the idea into functional and technical requirements.

2

Phase Objective

Translate the idea into functional and technical requirements.

- User inputs: name/user ID, age, weight, fitness goal and workout intensity.
- Functional requirements: generate a 7-day plan, provide nutrition/recovery tips, accept feedback and update plans.
- Admin requirement: view registered users and original/updated plans.
- Technology requirements: Python, FastAPI, Gemini API, Jinja2, SQLAlchemy, SQLite and Uvicorn.

Project Design Phase

Design the architecture, modules and user interaction flow.

3

Phase Objective

Design the architecture, modules and user interaction flow.

- Frontend: index.html, result.html and all_users.html using Jinja2 templates.
- Backend: FastAPI routes.py handles input, routing and AI calls.
- AI layer: Gemini 1.5 Pro for workout generation/update; Gemini Flash for nutrition tips.
- Data layer: SQLite + SQLAlchemy stores user details and both original and updated plans.

Project Planning Phase

Break implementation into milestones and activities.

4

Phase Objective

Break implementation into milestones and activities.

- Milestone 1: model selection, architecture definition and environment setup.
- Milestone 2: core functionality and FastAPI backend development.
- Milestone 3: routes.py application logic.
- Milestone 4: frontend development with HTML/CSS/Jinja2.
- Milestone 5: local deployment and deployment verification.

Project Development Phase

Implement the AI workflow, backend, database and interface.

5

Phase Objective

Implement the AI workflow, backend, database and interface.

- `generate_workout_gemini()` creates structured 7-day workout plans.
- `generate_nutrition_tip_with_flash()` provides concise goal-aligned advice.
- `update_workout_plan()` revises plans using user feedback.
- FastAPI routes connect forms, Gemini functions, database operations and templates.
- SQLite/SQLAlchemy persists user information and plan versions.

Project Testing Phase

Verify functionality and local deployment.

6

Phase Objective

Verify functionality and local deployment.

- Run the FastAPI server with Uvicorn and access the application locally.
- Test the main workflow: submit user details → generate plan → display result.
- Test feedback workflow: submit feedback → regenerate plan → store updated plan.
- Verify nutrition-tip generation and admin user/plan view.
- Use FastAPI /docs for interactive API endpoint testing.

Project Documentation Phase

Record the implementation details, workflow and project outcomes.

7

Phase Objective

Record the implementation details, workflow and project outcomes.

- Document project description, scenarios and technical architecture.
- Document milestones, activities, functions, routes and templates.
- Record Gemini model selection and reasons for integration.
- Describe database storage, frontend pages, deployment steps and user flows.
- Maintain a clear record of original and updated plan behavior.

Project Demonstration Phase

Demonstrate the complete FitBuddy user journey.

8

Phase Objective

Demonstrate the complete FitBuddy user journey.

- 1. Open the FitBuddy homepage and enter user details.
- 2. Generate a personalized 7-day workout plan.
- 3. Show the nutrition/recovery tip.
- 4. Submit feedback such as adding cardio or rest days.
- 5. Demonstrate the updated plan and confirmation.
- 6. Show the admin view containing users and original/updated plans.

FitBuddy System Architecture

High-level flow derived from the project architecture and routes



Core AI flow: user input → FastAPI → Gemini generation → result page → feedback → Gemini update → database

Technology Stack

Technologies explicitly documented in the project

Python

Application programming

FastAPI

Backend + API routing

Gemini 1.5 Pro

Workout generation + updates

Gemini Flash

Fast nutrition tips

Jinja2 + HTML/CSS

Dynamic frontend

SQLite + SQLAlchemy

Persistent data storage

Uvicorn

Local ASGI server

Git

Version control prerequisite

Conclusion

Project outcome and key contribution

0

FitBuddy Outcome

An adaptive AI-powered fitness planning web application

- Generates personalized 7-day workout plans from user inputs.
- Provides goal-aligned nutrition or recovery tips.
- Uses feedback to dynamically revise workout plans.
- Stores users and plan versions for later access.
- Provides an admin view for monitoring users and plans.
- Demonstrates integration of generative AI with a FastAPI web application.